MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISSION ONWARDS)

MG4DSEAQC202 – **Physiology and Internal Organisation of Finfishes and Shellfishes**

Duration: 1.5 hrs Maximum Marks: 50

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A
Very Short Answer Questions
Answer any 10 out of 10
Each question carries 1 mark

1. Explain Crystalline Style? [1] [U]

2. Distinguish between Holobranch and Hemibranch? [1] [U]

3. Distinguish between Physostomous and Physoclistous swim bladder in fishes? [2,4] [U]

4. Name the three otoliths found in the internal ear of fishes? [3][U]

5. What are Ommatidia? [3][U]

6. Distinguish between Strainers and Grazers in fish feeding habits, with examples? [4] [U]

7. Distinguish between Isometric and Allometric growth in fishes? [5][U]

8. Distinguish between Relative and absolute fecundity? [5][U]

9. What are Pyloric Caecae? [1] [U]

10. What is Appendix Masculina? [6][U]

[1 x 10=10] Marks

Part B
Short Answer Questions
Answer 5 Questions
Each question carries 6 marks

11. Explain briefly any six Accessory Respiratory Organs in fishes [1] [U]

12. Describe the Arterial System in Teleosts with the help of a suitable diagram? [1][U]

13. Explain Migration in fishes, with suitable examples? [2,4][U]

14. Compare Osmoregulation in Freshwater and Marine Teleosts [3][U]

15. Explain the organization and functions of Lateral Line System in fishes, with suitable diagrams [3] [U]

16. Elaborate upon Sexual Dimorphism in fishes with suitable examples [6] [A]

17.Explain the different Indirect methods for the estimation of growth in fishes? [5] [An]

18. Illustrate the structure of gills and methods of feeding in bivalves [4] [U]

[6 x 5 = 30] Marks

Part C
Essay Type Questions
Answer any one Question
Each question carries 10 marks

19Explain the organs involved and physiology of Excretion in shrimps and Bivalves? [3] [U]

20. Describe the structure of heart and circulatory system of bivalves, with suitable diagrams [1][U]

21. Elaborate upon different methods for studying reproduction and reproductive seasons in fishes? [5][An]

[10 x 1 = 10]Marks